DUBAI MUNICIPALITY AI PARK DESIGN CHALLENGE

One-minute Concept Animation

Storyboard and supporting documentation for the 60-second film.

12

OF 12 UPLOAD SLOTS

144 m	131	2,595 m²	64.6%
CRESCENT ARC LENGTH	TREES PLANTED	CANOPY AREA	COMFORTABLE AS DESIGNED

▶ **WATCH THE 60-SECOND CONCEPT ANIMATION FILM ONLINE (4K)**

wasim437.github.io/AI_SAFA/submission/12_Concept_Animation_Video/concept_film_hero.html

VISIT THE LIVE PROJECT PORTAL: wasim437.github.io/AI_SAFA/

▶ **VISIT THE LIVE PROJECT PORTAL**

wasim437.github.io/AI_SAFA/

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01 WHAT IS IN THIS FILE

Concept Animation Storyboard	DRAWING
Fig10 masterplan	PLAN

02 THE SCALE OF THE ANALYSIS BEHIND IT

39 years of climate normals (NCM, 1977–2015) rebuilt into an 8,760-hour modelled year

8,760 hours of sun position ray-traced against 131 trees for every hour of the year

15,000 one-metre ground cells modelled for shade and comfort, site-wide

4 models trained and tested for leakage — two of them decide what this document reports

41 checks run on every rebuild, so a wrong number fails loudly instead of shipping quietly

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository	github.com/wasim437/AI_SAFA
Live portal	wasim437.github.io/AI_SAFA/
Produced by	**BUILD** tools/sync_film.py **TESTS** tests/test_film.js
Reproduce	`python run_analysis.py` · `python -m tests.test_pipeline`

Generated 2026-08-06 · reproducible with python tools/build_submission_pdfs.py

A ten-phase process, checked at every step

Nothing downstream is hand-drawn or hand-typed — change an input and every figure moves with it, or a test fails loudly.

01 THE TEN PHASES

P1 Site & Context Analysis
39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

P2 Problem Definition
Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

P3 Opportunity & Objectives
The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

P4 Concept Development
Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

P5 Masterplan Development
Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

P6 Detailed Design
The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

P7 Performance & Sustainability
Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

P8 User Experience & Activation
K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

P9 **AI Workflow & Visualisation** THIS DOCUMENT
The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

P10 Submission Assembly
Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

02 THE CODE AND DATA BEHIND THIS DOCUMENT

Sync film
tools

Test film
tests

03 REPRODUCE IT END TO END

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 41 checks
```

UPLOAD SLOT 12 OF 12 · ONE-MINUTE CONCEPT ANIMATION — STORYBOARD

Concept Animation — Storyboard

Sixty seconds, drawn from the project's own analysis

The film is not an illustration of the design; it is rendered from the design. Its geometry comes from the same arc definition as the masterplan and its sun from the same 8,760-hour solar model, so it cannot drift away from the drawings the way an animation produced separately would.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 05 August 2026 · every quantity regenerated by `python run_analysis.py`

1. Structure — five scenes, sixty seconds

Time	Scene	What it shows
0–7 s	**The site**	Al Safa 2 as it is today: flat, exposed, and empty at midday
7–18 s	**The heat**	The reason it is empty — the modelled heat index across a summer day, and the hours that are lost to it
18–38 s	**The crescent**	The arc builds and plants itself while a full computed day passes over it. 131 trees, the falaj, the radial alleys and the rooms appear in the order they are reached
38–48 s	**Beneath it**	Eye level on the walk: the dappled soffit, the louvre cutting the low sun, the channel at the edge
48–60 s	**The proof**	The measured result — comfortable hours rising from 44.5% to 64.6% — as the park fills at dusk

Scene boundaries are asserted by tests/test_film.js, which renders every frame and fails on any gap, overlap or invalid value.

2. Why it can be trusted against the drawings

- The arc is the same 141 m radius as the masterplan, regenerated by tools/sync_film.py from src/plan.py.
- The sun positions are the same NREL-computed values used by the shade model — the shadows move correctly for the site's latitude and the time of year being shown.
- The tree count, the 0.9 m channel and the room layout are read from the same geometry as every drawing in this submission.
- Every frame is rendered by an automated test that fails on any NaN or undefined value, so a geometry change cannot break the film quietly.

3. Three cuts, one film

The submitted video is the hero cut. Two others exist because they answer different questions, and all three run sixty seconds against the same narration and the same figures.

Cut	What it shows	Built by
concept_film_hero.html the submitted film	Drawn entirely from the project's data — no photographs. Seven plan forms swept against the 8,760-hour solar model and scored, a real August day passing over the adopted arc, and a peak heat index falling from 56.8 °C to 48.7 °C.	tools/build_concept_film_hero.py
concept_film_presentation.html	The six photoreal renders, moving between each visualisation and the analysis behind it.	tools/build_concept_film_v2.py
concept_film.html	The park drawn in 3D in real time from src/plan.py with shadows computed live — evidence that the film and the drawings share one geometry.	hand-authored; geometry by tools/sync_film.py

All three record their own MP4. Every frame of the third is asserted by tests/test_film.js.

The hero cut is submitted because the plan-form sweep is the part of this proposal that cannot be photographed. The straight bar shades more ground on average and the film shows it doing so; the arc is adopted on hours in the year when the route offers nowhere at all to stand, 330 down to 52. That is the clearest available answer to how AI changed the design rather than decorated it.

3. The narration

The film carries a spoken commentary in four fifteen-second segments, cued to 0, 15, 30 and 45 seconds. The voice is synthesised; the words are not. Every figure it speaks — 56.8 °C in the open, a 141 m radius, 44.5% rising to 64.6% — is the same figure the analysis produces and the same one printed elsewhere in this submission.

The audio is embedded in the page rather than kept in files beside it. Opened as a local file, a browser refuses to load a sibling audio file at all, and cannot route one into a recording; inlined, it does both. That also keeps the film a single standalone file, which is the same reason its geometry is inlined rather than fetched.

4. How to produce the video file

The film records itself. Every frame is a pure function of one number — the time in seconds — so the page can hand the canvas and the narration straight to the browser's recorder and write the video out. Nothing is screen-captured, so the file carries no browser chrome, no desktop behind it, and no dependence on what the window was scaled to.

- Open `concept_film.html` in Chrome or Edge.
- Press **Record the film to a video file** and leave the tab in front for sixty seconds.
- **Quality** beside it switches the recording between 1920×1080 and 3840×2160. At 4K the film's backing canvas is doubled and the drawing context scaled, so every line and glyph is resolved at 4K rather than upscaled from 1080p — the bitrate rises with it.
- The file downloads on its own as `Falaj_Al_Safa_Concept_Film_60s_4K.mp4` — sixty seconds with narration, about 43 MB at 4K or 28 MB at 1080p.
- **Narration on / off** beside it records a silent version instead, if a voice-over is not wanted.

Browsers that cannot write MP4 fall back to WebM, and a WebM written this way carries no duration in its header — it plays, but some players show no timeline. Convert it with `ffmpeg -i film.webm -c:v libx264 -pix_fmt yuv420p film.mp4`, or record again in Chrome or Edge, which write MP4 directly.

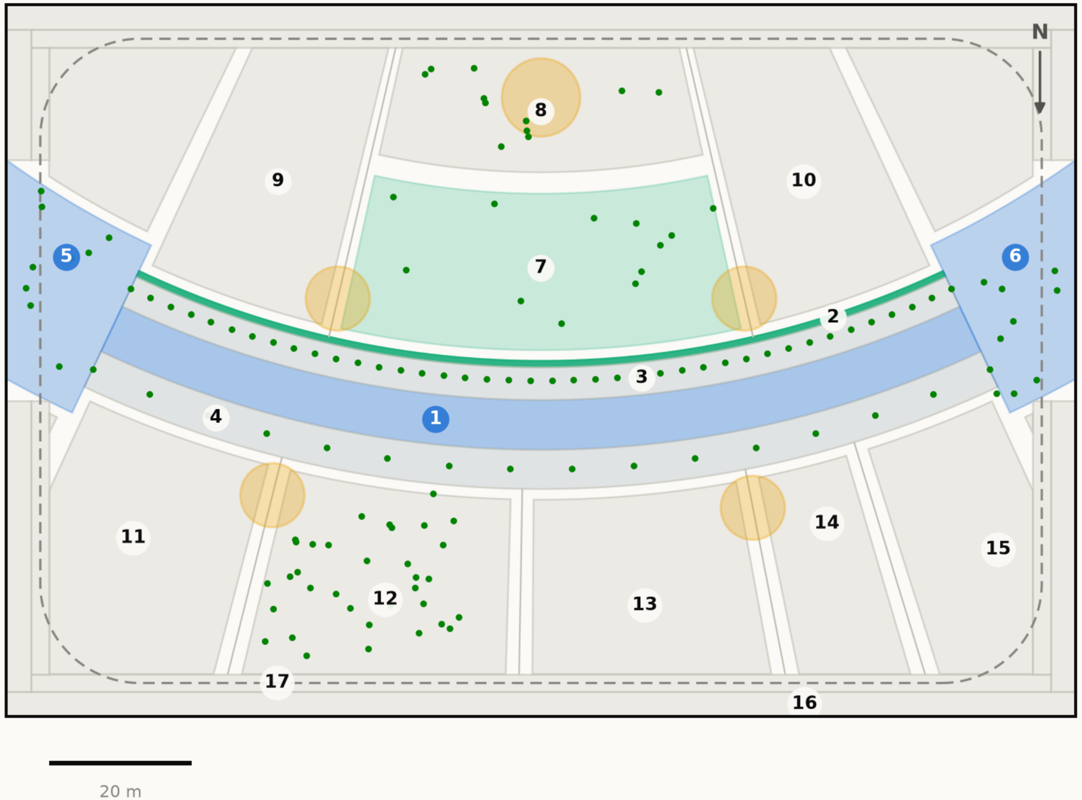
The film shows the **analysed** scheme exactly: the same arc, the same trees, the same solar model. It is therefore consistent with every number quoted elsewhere in this submission.

Fig10 masterplan

The plan as the code draws it — every room struck off the arc centre, areas measured from the drawn polygon

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
	Al Sikkak — shaded alleys & setbacks	1,414 m²
	Total	**15,000 m²**

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

DRAWN BY src/plan.py src/figures.py **READ FROM** data/raw/site_zoning_schedule.csv data/processed/masterplan_geometry.json

Every path above is a live link. Nothing on this sheet was drawn by hand — regenerate it with python run_analysis.py

Every step of the work, with its proof attached

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out. Every file name is a live link — click it and read the actual thing, do not take this document's word for it.

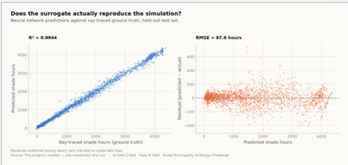


fig05_surrogate_performance.png

PHASE 9 AI Workflow & Visualisation

THIS DOCUMENT RESTS ON IT

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

CODE src/models.py src/boards.py tools/sync_film.py

PROOF — click to open models/model_metrics.json tests/test_pipeline.py

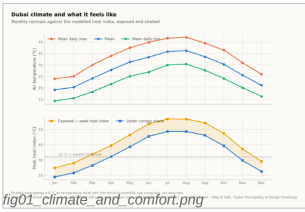


fig01_climate_and_comfort.png

PHASE 1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

CODE src/climate.py src/solar.py

PROOF — click to open data/processed/hourly_climate_comfort_8760.csv data/raw/sources.json

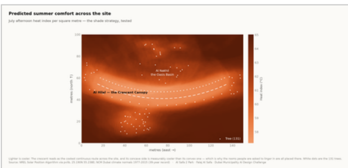


fig04_site_comfort_map.png

PHASE 2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

CODE src/climate.py

PROOF — click to open models/headline_metrics.json

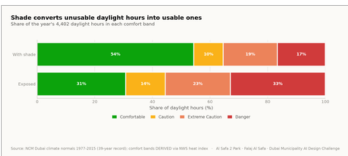


fig02_comfort_bands.png

PHASE 3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

CODE src/config.py src/plan.py

PROOF — click to open models/headline_metrics.json

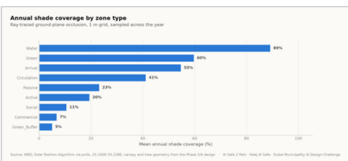


fig03_shade_by_zone.png

PHASE 4 Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

CODE src/plan.py src/solar.py

PROOF — click to open data/processed/masterplan_geometry.json

WHAT WE DID, AND HOW TO CHECK IT

The work, with its proof attached — continued

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out. Every file name is a live link — click it and read the actual thing, do not take this document's word for it.

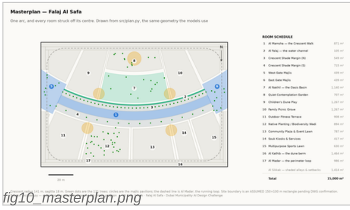


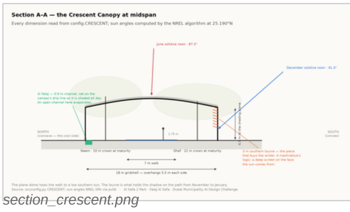
fig10_masterplan.png

PHASE 5 Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

CODE src/plan.py src/figures.py

PROOF — click to open data/raw/site_zoning_schedule.csv data/processed/masterplan_geometry.json



section_crescent.png

PHASE 6 Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

CODE src/drawings.py src/config.py

PROOF — click to open data/processed/planting_layout.csv

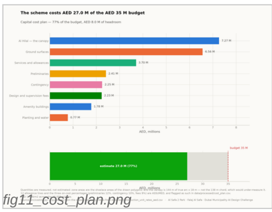


fig11_cost_plan.png

PHASE 7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

CODE src/costing.py src/climate.py

PROOF — click to open data/processed/cost_plan.csv models/cost_summary.json

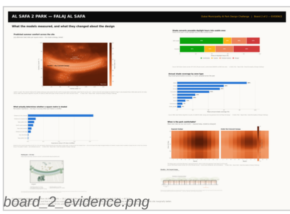
fig08_microclimate_regimes.png

PHASE 8 User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

CODE src/models.py src/dataset.py

PROOF — click to open data/processed/spatial_grid_comfort.csv



board_2_evidence.png

PHASE 10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

CODE tools/build_submission_pdfs.py tools/build_reports.py

PROOF — click to open tests/test_pipeline.py

THE VISUAL RECORD

The 1 image in this file, and the whole record

12

OF 12

This slot's own pictures come first, larger. The remaining 23 of the 24 images the pipeline produces follow, so the whole visual record is in every file. Each links to its full-resolution original; nothing here is hand-drawn.

01 IN THIS FILE

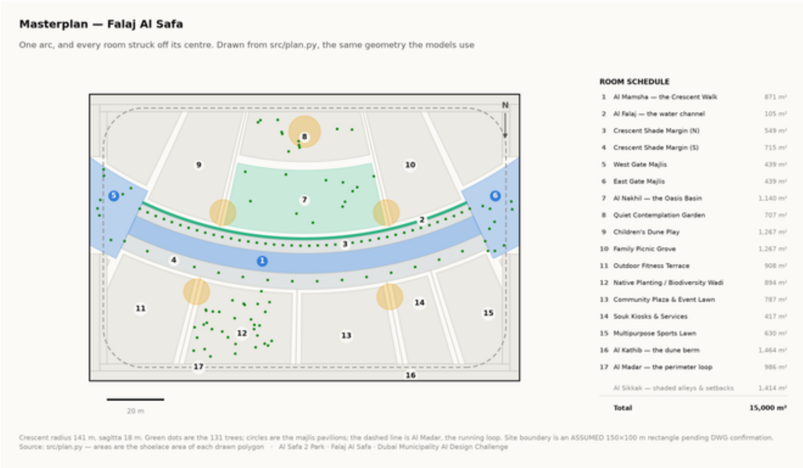


Fig10 masterplan
analysis output

02 EVERY OTHER IMAGE IN THE PROJECT

The other 23 images the project produces are not reprinted here — this file shows its own. Each name below opens the full-resolution original in the repository.

Fig01 climate and comfort *analysis output*
Fig02 comfort bands *analysis output*
Fig03 shade by zone *analysis output*
Fig04 site comfort map *analysis output*
Fig05 surrogate performance *analysis output*
Fig06 feature importance *analysis output*
Fig07 confusion matrix *analysis output*
Fig08 microclimate regimes *analysis output*
Fig09 diurnal comfort *analysis output*
Fig11 cost plan *analysis output*
Circulation *technical drawing*
Elevation *technical drawing*
Facilities *technical drawing*
Planting *technical drawing*
Section *technical drawing*
Board 1 concept *presentation board*
Board 2 evidence *presentation board*
Masterplan aerial golden hour *artistic impression*
Oasis basin *artistic impression*
Spine corridor interior *artistic impression*
Night plaza render *artistic impression*
Childrens dune play *artistic impression*
Souk plaza *artistic impression*

Proof the analysis runs, and what it reports

These are the values the pipeline wrote on its last run, read straight out of the files it produced. The commands below regenerate every one of them from the published repository.

THE FIGURES THIS FILE LEADS ON

each one appears in the full tables below, from the same file

144 m CRESCENT ARC LENGTH	**131** TREES PLANTED	**2,595 m²** CANOPY AREA	**64.6%** COMFORTABLE AS DESIGNED

MEASURED RESULT

models/headline_metrics.json

Annual daylight hours modelled	**4,402**
Comfortable daylight hours — today	**44.5%**
Comfortable daylight hours — as designed	**64.6%**
Mean heat-index reduction under canopy	**7.13 °C**
Peak heat index, exposed	**56.8 °C**
Peak heat index, shaded	**48.7 °C**
Crescent Walk shaded, canopy and louvre	**87.3%**
Site-wide mean shade	**34.1%**
Trees planted	**131**

TRAINED MODELS, ON HELD-OUT TEST SETS

models/model_metrics.json

M1a Random Forest — shade surrogate, test R^2	**0.9982**
M1b Neural network — deployed surrogate, test R^2	**0.9944**
M2 Gradient Boosting — comfort band, test accuracy	**97.5%**
M3 K-Means — regimes selected by silhouette	**k = 2**
Training / validation / test split	**10,500 / 2,250 / 2,250**
Random seed, fixed for reproducibility	**42**

WHAT THE CHECKS PROTECT

each one asserts what would otherwise drift silently

Climate fidelity
the modelled year reproduces the published normals it was built from

No overlaps
no room in the schedule overlaps another

Areas close
every area sums back to 15,000 m²

Budget held
the cost plan stays inside AED 35 M

No leakage
no model can see its own answer in its inputs

RUN IT YOURSELF

```
git clone https://github.com/wasim437/Al_SAFA.git
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m tests.test_pipeline    # 41 correctness checks
node docs/_PORTAL/selftest.js    # 64 portal checks
node tests/test_film.js          # every frame of the film
```

The complete project, and where to check it

Every one of the 125 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository	**github.com/wasim437/AI_SAFA**
Live portal	**wasim437.github.io/AI_SAFA/**

THE SOURCES BEHIND THIS FILE *open these first if you want to check this document*

tools/sync_film.py	Technical drawing — to scale.
tests/test_film.js	Technical drawing — to scale.

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Ma…
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Di…
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (6)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
README.md	The design argument, written for a juror
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/Al_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data
figures/fig05_surrogate_performance.png	Analysis output — computed from project data

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

The complete project — continued

figures/fig06_feature_importance.png	Analysis output — computed from project data
figures/fig07_confusion_matrix.png	Analysis output — computed from project data
figures/fig08_microclimate_regimes.png	Analysis output — computed from project data
figures/fig09_diurnal_comfort.png	Analysis output — computed from project data
figures/fig10_masterplan.png	Analysis output — computed from project data
figures/fig11_cost_plan.png	Analysis output — computed from project data

THE TECHNICAL DRAWINGS AND BOARDS (7)

design/visuals/circulation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/elevation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/facilities_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/planting_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/section_crescent.png	Technical drawing — generated from src/plan.py
design/boards/board_1_concept.png	Technical drawing — generated from src/plan.py
design/boards/board_2_evidence.png	Technical drawing — generated from src/plan.py

THE ANALYSIS CODE (13)

src/__init__.py	Analysis module
src/boards.py	The two presentation boards
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/config.py	Every constant the project reads
src/costing.py	The cost model against the AED 35 M ceiling
src/dataset.py	Assembles the ML training tables
src/drawings.py	Section, elevation, circulation, planting, facilities
src/figures.py	The analysis charts
src/models.py	The four models
src/plan.py	SINGLE SOURCE of the crescent geometry
src/solar.py	Sun position and shadow ray-tracing
src/viz.py	Analysis module
run_analysis.py	Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (17)

- tools/__init__.py
- tools/build_concept_film_hero.py
- tools/build_concept_film_v2.py
- tools/build_docs.py
- tools/build_links.py
- tools/build_notebook.py
- tools/build_reports.py
- tools/build_submission_pdfs.py
- tools/check_renders.py
- tools/embed_narration.py
- tools/make_video_mp4.py
- tools/make_winning_video.py
- tools/organize_repo.py
- tools/report_content.py
- tools/sync_film.py
- tools/sync_portal.py
- tools/sync_submission.py

THE DATA, AS ISSUED (7)

data/raw/construction_unit_rates_aed.csv	Source dataset
data/raw/dubai_climate_normals_ncm.csv	Source dataset
data/raw/dubai_population_dsc_2023.csv	Source dataset

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The complete project — continued

data/raw/site_zoning_schedule.csv	The measured room schedule
data/raw/sources.json	Every source dataset, with its period
data/raw/species_water_carbon_rates.csv	Source dataset
data/raw/walk_catchment_rings.csv	Source dataset

THE DATA, AS PROCESSED (6)

data/processed/cost_plan.csv	The capital cost plan, line by line
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/masterplan_geometry.json	The plan geometry, as exported
data/processed/planting_layout.csv	Every one of the 131 trees
data/processed/schema.json	Generated by the pipeline
data/processed/spatial_grid_comfort.csv	The 15,000-cell ground grid

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	41 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded

THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Al Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD_PARKS_MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file

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The complete project — continued

submission/09_Material_Landscape_Palette/MANIFEST.md	What is in the slot, and what produced each file
submission/10_Complete_Design_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md	What is in the slot, and what produced each file
submission/12_Concept_Animation_Video/MANIFEST.md	What is in the slot, and what produced each file

WORKING HISTORY (1)

archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why

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